CLT-Det: Correlation Learning Based on Transformer for Detecting Dense Objects in Remote Sensing Images

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***Abstract*— Challenges still exist in the task of object detection in remote sensing images with densely distributed objects due to large variation in scale and neglect of the relative position and correlation. To address these issues, a correlation learning detector based on transformer (CLT-Det) is proposed for detecting dense objects in remote sensing images. A transformer attention module (TAM) is designed to improve the densely packed objects' model representation ability by learning pixelwise attention with a transformer. To alleviate the semantic gap caused by the variations in scale, a feature refinement module (FRM) is proposed by improving the multiscale feature pyramid. A correlation transformer module (CTM) is proposed to extract correlation information and it encodes position information of dense objects' features on the classification branch for fully using the position information and correlation among objects. Extensive experiments compared with several state-of-art methods on two challenging remote sensing datasets, namely, dataset for object detection in aerial images (DOTA) and HRSC2016, demonstrate that the proposed CLT-Det achieves promising and competitive performance.**

***Index Terms*— Correlation learning, deep learning, object detection, remote sensing images, vision transformer (ViT).**

I. Introduction

BENEFITING from the wide application of convolutional neural networks (CNNs) in computer vision, object detection has developed rapidly in recent years. Most of the existing detectors [1], [2], [3] aim to detect objects by horizontal bounding boxes (HBBs) in natural images and achieve considerable progress. Whereas in remote sensing images, the objects are densely distributed in a clustered environment, each object has an arbitrary orientation in the overhead view, and there also exist large-scale discrepancies among different instances. General CNN-based detectors are not suitable to solving the problems of dense distribution and scale variation, thus making them difficult to be widely applied in remote sensing images. Consequently, localization and classification of dense objects from information-rich remote sensing images are both challenging and significant.

During the early period, the general CNN-based detectors are directly used for object detection in remote sensing images because of the replacement of handcrafted features and the powerful classification and localization abilities. However, due to the large resolution of remote sensing images, convolution is time-consuming and computationally intensive, causing feature extraction to be ineffective. Therefore, some methods crop the high-resolution images into image patches, and then the patches are detected by a general detector. These methods significantly improve the performance of the CNN-based algorithms for object detection in remote sensing images. Subsequently, since the objects in the remote sensing images are all in arbitrary orientations, more works [4], [5], [6], [7], [8], [9], [10] have explored the use oriented bounding boxes (OBBs) to locate densely arranged objects and have achieved promising results on several challenging remote sensing image datasets. Most of the recent methods primarily concentrate on addressing the representation problem of OBBs [7], [11], [12], [13], the boundary problem of OBBs [14], [15], [16], and rotation invariance in feature extraction [17], [18], [19], [20], but ignore the correlation learning of dense objects.

In practice, due to the bird's-eye views of remote sensing images, a massive number of objects are often distributed intensively and arranged in a cluttered manner. As shown in Fig. 1, the category and location of one object can usually be determined by the presence of adjacent or other objects in the same scene. Thus, the correlation information is essential. Furthermore, natural images also have the issue of large-scale variations, which is addressed using multilevel feature pyramid network (FPN) [21] in general methods. However, the variations in object scales within remote sensing images are more drastic. Some small objects (vehicles, etc.) occupy only

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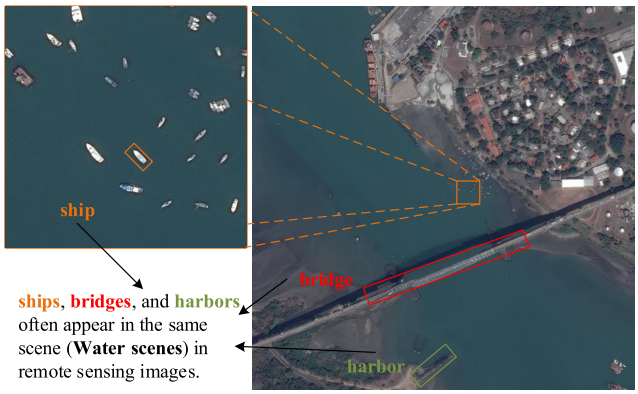


Fig. 1. Sample images from the DOTA dataset show that multiple objects (SH, BR, and HA) in a single image are correlated.

a few pixels, while others (harbor (HA), stadiums, etc.) occupy hundreds or thousands of pixels. Such extreme scale variation would lead to a huge semantic gap across the individual levels of the feature pyramid.

In this article, we introduce a transformer into the networks and refine the FPN to address these challenges. We propose a novel correlation learning detector based on transformer (CLT-Det) tailored for dense and multiscale objects in remote sensing images. It consists of three modules optimized for the detector backbone network, the FPN, and the head network: 1) a transformer attention module (TAM); 2) a feature refinement module (FRM); and 3) a correlation transformer module (CTM).

First, the TAM is proposed to extract robust features from dense object by leveraging the structure of the transformer which can capture the correlation of semantic information cross-channel to build dense attention at the pixel level. Second, considering that the objects have a large-scale variation, FRM is proposed to optimize the multiscale feature pyramid. The proposed FRM refines the semantic information of the top layer features which contains the richest semantic information and guides the feature learning at each level through channels to alleviate the semantic gap among features at different scales. Finally, inspired by the transformer-based image captioning model [22], [23], [24], we propose CTM for correlation learning of dense objects. As mentioned earlier, the categories of dense objects are difficult to be judged in a cluttered context. We capture the correlation of these objects with the transformer and introduce the learned correlation features to the classification branch of the detector head network to assist the model in classifying dense objects accurately. In summary, the main contributions of this article are as follows.

1) The TAM is proposed to improve the model's capability to represent dense objects by capturing the correlation among different features.
2) The FRM is proposed to avoid the semantic gap due to scale variations by propagating the semantic information of top layer features to other levels by channel guidance.
3) We propose a CTM to address the challenge of dense object classification. The CTM can introduce the location information of each object to assist the classification of dense objects by joining it to the head network of the detector.
4) Extensive experiments are conducted on the dataset for object detection in aerial images (DOTA) and HRSC2016 datasets to prove the effectiveness of our proposed methods.

II. Related Work

A. Vision Transformer

Transformer [25], which discarded the traditional convolutional and recurrent neural networks, was proposed to use the self-attention and feed-forward layers for natural language processing tasks such as machine translation and has achieved promising performance. Due to its simple structure and powerful relationship capturing capability, the transformer was applied to image captioning and made good progress as well recently. In image captioning [22], [23], [24], a general detector extracted and saved multiobjects' features which were input to the transformer after RoIPooling [1]. The features were encoded and decoded to output captions finally.

Recently, the transformer as a CNN alternative for vision tasks has shown great potential and considerable progress. Vision transformer (ViT) [26] was first proposed to use transformer in vision tasks and provided a new paradigm for subsequent research. It divided an image into multiple patches which were regarded as the word tokens and introduced position encoding according to the position of the patches, which was finally input to the transformer and output classification confidence after encoding. Data-efficient image transformer (DeiT) [27] applied the distillation strategy to study the data efficiency problem in ViT with comparable results on ImageNet. Instead of a fully transformer structure, the combination of transformer and CNNs has recently been a mainstream research direction. Convolutional vision transformers (CvT) [28] was proposed to use the translation invariance and scale invariance of convolution to improve the performance and efficiency of ViT. For the sake of less additional training data and supervision information, Yuan *et al.* [29] combined CNN with ViT to strengthen the locality by introducing convolution from the input, feed-forward layers, and the top of the transformer. Moreover, pyramid vision transformer (PvT) [30] was proposed for dense prediction tasks, fusing the feature pyramid structure with the transformer for better output of multiscale features. A Swin transformer [31] was proposed to construct multiscale features using a hierarchical method and the local properties of convolution. Several studies [32], [33], [34], [35] have enabled the transformer to achieve results beyond CNNs in visual tasks.

In CLT-Det, the transformer is used to capture the dependencies among multichannel features in feature extraction and to obtain pixelwise attention of dense objects. In addition, we also exploit the transformer's ability to model multiobjects' features in image captioning to explore the effect of multiobjects' correlations on improving the performance of multiobjects' classification.

B. Multiscale Feature Learning

To tackle the issue of scale variation in objects in images, FPN [21] was the first to propose the structure of a multilevel

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feature pyramid to learn multiscale features and make multiscale predictions through a top-down path. Path aggregation network (PANet) [36] extended the FPN with a new bottom-up path that leveraged the low-level features of the network for prediction. Libra region-CNN (R-CNN) [37] was proposed as a balanced feature pyramid, which used semantic information of higher level features and detailed information of lower level features to improve the object detection accuracy complementarily. To search for a better FPN structure, NAS-FPN [38] was proposed to use neural architecture search (NAS) to find a better way of feature extraction in the search space. Bidirectional FPN (BiFPN) [39] was proposed to handle multiscale features equally by learning different weights for different levels of features. AugFPN [40] proposed a consistent supervised approach to improve the problem of semantic gaps in feature fusion by computing candidate regions and losses separately before and after feature fusion. However, AugFPN triggered a lot of extra computations during training. Qiao *et al.* [41] proposed the structure of a recursive FPN by adding extra feedback connection to achieve two or more feature extractions for an image. Laplacian FPN (L-FPN) [42] used the Laplace operators to obtain high-frequency features and low-frequency features and fused them to obtain multiscale target features.

The above methods effectively improved the capability of feature pyramid for multiscale objects feature extraction but still could not confront the drastic scale variations in objects in remote sensing images, due to the semantic gap triggered by information loss on multiscale features. Based on the methods above, we aggregate semantic information from top-level features to cope with the scale variations in objects in remote sensing images [43].

C. Object Detection in Remote Sensing Images

With the success of the general detector R-CNN [44] in natural scenes, most of the R-CNN-based object detection algorithms were applied in remote sensing image object detection. Xia *et al.* [45] detected the large-scale remote sensing dataset DOTA by the general detector Faster R-CNN [1] and achieved acceptable results. With regard to the existing CNN architecture, Cheng *et al.* [46] proposed a network to improve the rotational invariance of the feature by rotating the training samples. Long *et al.* [47] presented an unsupervised score bounding box regression combined with a nonmaximum suppression (NMS) algorithm to optimize the localization ability of the detector. Pang *et al.* [43] introduced global attention to increasing the receptive field and contextual information of feature extraction to suppress false detection in the model, which effectively improved the detection performance. Furthermore, you can only look twice (YOLT) [48] introduced feature fusion and modified the stride size of the convolution on the general detector YOLO [49], aimed to be more adapted to the detection of small objects in remote sensing images. Refined single-shot multibox detector (RSSD) [50] proposed an accurate and fast detector using different variations in enhanced samples to solve the problem of false positives.

The methods above were all based on HBBs. However, most of the objects were densely distributed with arbitrary orientations in remote sensing images. The OBBs' regression is a main research direction of the current object detection methods for remote sensing images. Initially, OBB was applied to optical character recognition (OCR) to make good progress [51], [52], [53]. Based on the above, Ma *et al.* [54] proposed a method to acquire more rotation-invariant features of objects by rotating the anchor. This method is widely used in text recognition and provided a baseline model for OBB-based object detection in remote sensing images. Subsequently, detection for small, cluttered and rotated objects (SCRDet) [5] was proposed for robust detection of objects in remote sensing images, and a loss function was designed to address the regression boundary problem of OBBs. Yang *et al.* [17] presented a single-stage detector for remote sensing images with large object scale aspect, and feature reconstruction was conducted by a refinement module. To further address the rotational invariance of the bounding box, Ding *et al.* [6] designed a learnable module that converts the horizontal region of interest (HRoI) into a rotational region of interest (RRoI), alleviating the misalignment problem between the RoIs and the objects. Context-aware detection network (CAD-Net) [55] devised a model that incorporated global contextual information and pyramidal local contextual information to guide the network to concentrate on regions that were richer in semantic information. Li *et al.* [56] proposed to utilize feature fusion to enhance the object representation information and weaken the background information at the channel level and pixel level. Zand *et al.* [8] designed an anchor-free detector that does not require additional angle information and encoded the precise position and orientation of the objects during the feature extraction. Liu *et al.* [9] presented an aspect ratio weighted angle loss that assigned different penalty factors to different aspect ratio objects. Zhu *et al.* [57] proposed adaptive period embedding (APE) instead of the complex angle regression operation. Xu *et al.* [7] proposed a gliding vertex approach instead of angle regression to avoid the confusion problem of order points. Circular smooth label (CSL) [16] and densely coded labels (DCL) [58] address the complexity of angle regression and the boundary problem of angle regression. They simplified angle regression into a classification problem and achieved excellent results.

The methods mentioned above addressed the OBB representation problem, boundary problem and rotation-invariant feature extraction, but lacked research on dense object correlation mining. In this article, we further investigate the effect of dense object correlation learning on detection and classification in remote sensing images based on these OBB methods.

III. Methodology

In this article, we propose an end-to-end detector called CLT-Det in remote sensing images. We first outline our architecture of CLT-Det. Section III-B reviews several necessary details of the original transformer encoder. Section III-C describes specifically the details of TAM, Section III-D presents the information on our proposed FRM, and finally CTM is introduced in Section III-E.

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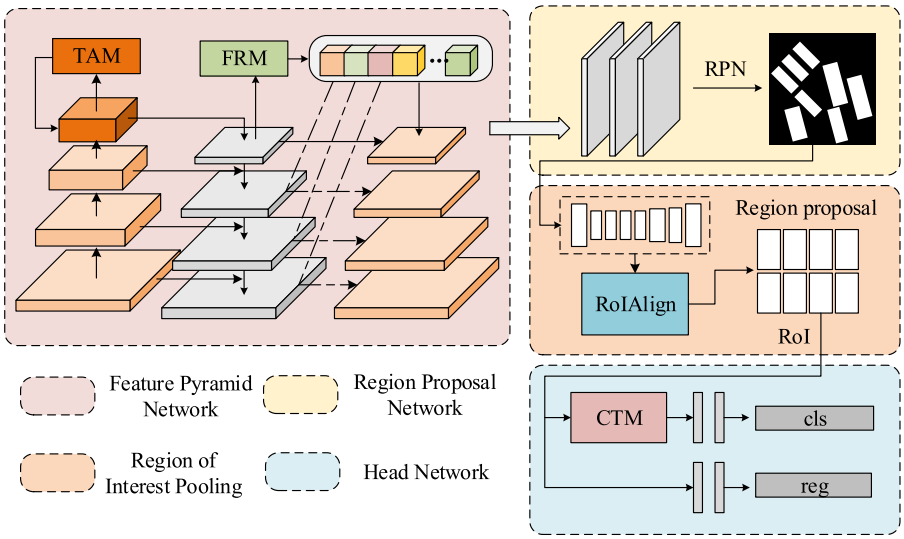


Fig. 2. Framework of the proposed CLT-Det. TAM is included in the feature extraction network and FRM is introduced in the feature pyramid. The correlation of objects is learned in the head network using CTM.

A. Overall Architecture of CLT-Det

Our CLT-Det is based on the two-stage object detection structure with the following main process.

1) Candidate Proposal Generation Stage: In this stage, residual network (ResNet) [59] is used as the backbone network of the entire detector for feature extraction. Particularly, the last stage features of ResNet are the most semantically informative, but limited by the scale to predict densely small objects. As shown in Fig. 2, the TAM module is introduced in the last stage of ResNet to strengthen the representation capability of dense objects. Subsequently, the feature pyramid is constructed using the extracted features of five stages as the original features at multiple scales. Unlike the traditional pyramid network, to deal with the drastic scale variations in remote sensing image objects, FRM is proposed to learn the semantic information of top-level features and guide the learning of features in other levels to prevent the semantic gap. Similar to other two-stage detectors, we construct region proposal network (RPN) at each level to produce candidate regions, and then obtain uniform features of the regions by region of interest alignment (RoIAlign) [60].

2) Localization and Classification Stage: In the second stage, we introduce CTM to enhance the classification capability of the head network for dense objects. The generated object locations and categories are processed by NMS to output the final results.

The whole network structure is exclusively tailored for remote sensing image objects, and the detectors are demonstrate to be more efficient and have better accuracy than other detectors.

B. Revisiting Transformer Encoder

The original transformer encoder is used to process serialized data, e.g., text and speech. As shown in Fig. 3, it consists of three major structures: 1) a multihead attention (MHA); 2) a feed-forward network (FFN); and 3) a residual structure.

The central module of the transformer is MHA, which is an arrangement of multiple self-attention models essentially.

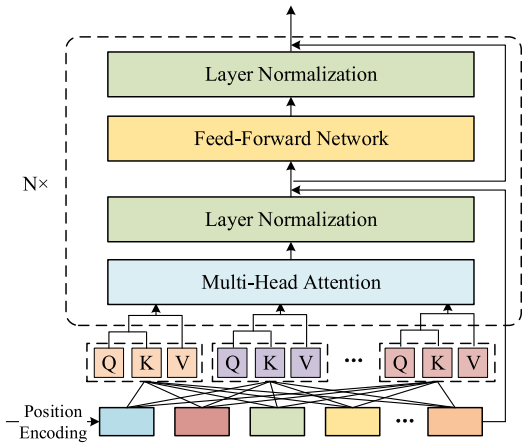


Fig. 3. Illustration of the transformer encoder block.

For self-attention, the input sequence embedding is first transformed into three matrices $Q \in \mathbb{R}^{n_q \times d_q}$, $K \in \mathbb{R}^{n_k \times d_k}$, and $V \in \mathbb{R}^{n_v \times d_v}$. Then attention is derived for Q, K, and V using scaled dot-product attention. Multiple self-attention layers are arranged together in MHA, and the attention weights in different feature spaces are output separately. The process of MHA can be formulated as follows:

$$\text{head}_i = \text{Self Attention}(Q_i, K_i, V_i) \tag{1}$$

$$\text{MHA}(Q, K, V) = \text{Concat}(\text{head}_i) \tag{2}$$

where head_i represents the $i_t h$ output of self-attention, and Concat means the concatenation operation. For the FFN layer, including the fully connected (FC) layer and the layer normalization, it is calculated as follows:

$$\text{FFN}(x) = \text{LN}(\text{MLP}(x)) \tag{3}$$

where x represents the output of MHA, LN represents the layer normalization layer, and MLP represents the multilayer perceptron. Eventually, the residual structure is introduced

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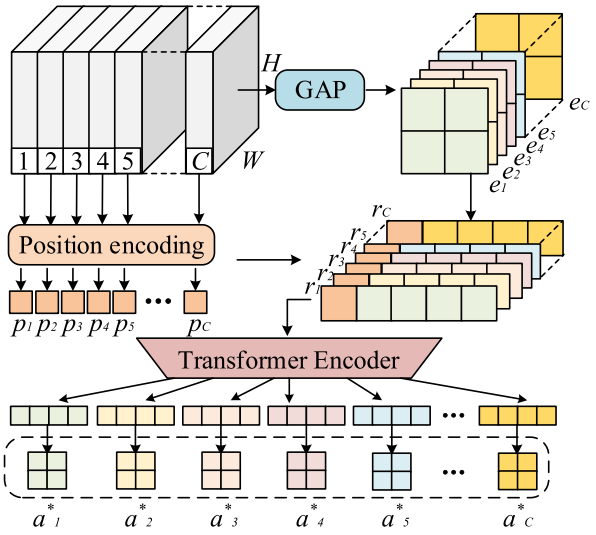


Fig. 4. Illustration of our proposed TAM.

among the layers of the transformer encoder, making the model training more stable.

C. Transformer Attention Module

The backbone network ResNet acquires semantic information-rich feature maps for subsequent detection after multiple convolutional layers. In remote sensing images, object features tend to be not prominent enough in the network due to object scale and arrangement limitations. We design a pixel-level attention network, namely, TAM, which is integrated into the backbone network ResNet and guides the network to enhance the representation capability of dense objects.

Intuitively, TAM is a module that combines channel-wise attention and pixelwise attention, ultimately achieving enhanced representation with pixelwise addition. As shown in Fig. 4, given the last stage of ResNet feature $\mathrm{X} \in \mathbb{R}^{C\times H\times W}$ where C, H, and W represent the channels, height, and width of feature maps, respectively, the feature embeddings are first obtained at the channel level by a global pooling operation (GAP). The ith feature embedding is calculated as follows:

$$e_i = \mathrm{GAP}_{k\times k}(f_i) \tag{4}$$

where f_i represents the ith channel feature, and k denotes the size after pooling (as shown in Fig. 4, $k = 2$). $\mathrm{GAP}_{k\times k}$ is used to preserve the spatial information of $k\times k$ features and provide the potential of subsequent implementation of pixelwise feature attention. Since the transformer is a serialized model, the feature embeddings are flattened as 1-D vectors as the input of the transformer. In this way, the feature map is decomposed into a sequence $[e_1, e_2, \ldots, e_C] \in \mathbb{R}^{C\times k^2}$, and each channel feature is considered as a "word embedding" in the original transformer. Furthermore, to introduce the positional relationship among channel features, the proposed method uses the position encoding to obtain each channel's position information, which is encoded to the feature embedding. The position encoding p_i^j is calculated as follows:

$$p_i^j = \begin{cases} \sin\left(i/10000^{j/512}\right) & \text{if} \quad j \bmod 2 == 0 \\ \cos\left(i/10000^{j/512}\right) & \text{else} \end{cases} \tag{5}$$

where p_i represents the location of the ith channel feature, and j represents the jth value in p_i. The obtained position information sequence $[p_1, p_2, \ldots, p_C] \in \mathbb{R}^{C\times k^2}$ is fused with the feature embedding sequence by summation and is fed to the transformer together to learn the correlation among features.

After each channel's feature embedding and location information is input to the transformer, a series of attention weights are initially learned. However, these weights only correspond to the interdependencies among individual channels and cannot learn attention information at the pixelwise, thus lacking the representation of dense objects. As shown in Fig. 4, to further obtain the dense attention, the proposed method reconstructs the transformer output weights which is $C \times k^2$ into a 2-D structure with each weight of size $k \times k$. The 2-D weights of $k \times k$ are not sufficient to guarantee attention to dense locations relative to the original features of $H \times W$, so the reconstructed weights are transformed into the dense attention of $H \times W$ by the upsample operation. The final attention weights $[a_1, a_2, \ldots, a_C] \in \mathbb{R}^{C\times H\times W}$ are output by a learnable convolutional layer, which are directly added to the original feature maps by a sigmoid function. The detailed weights are calculated as follows:

$$[r_1, r_2, \ldots, r_C] = [e_1, e_2, \ldots, e_C] + [p_1, p_2, \ldots, p_C] \tag{6}$$

$$\left[a_1^*, a_2^*, \ldots, a_3^*\right] = \mathrm{RCons}_{k\times k}(\mathrm{TE}([r_1, r_2, \ldots, r_C])) \tag{7}$$

$$[a_1, a_2, \ldots, a_C] = \sigma\left(\mathrm{Conv}_{1\times 1}\left(Up_{H\times W}\left(\left[a_1^*, a_2^*, \ldots, a_3^*\right]\right)\right)\right) \tag{8}$$

where r_i represents the ith input of the transformer, a_i^* represents the ith reconstructed weight, and a_i denotes the ith attention weight to add to the original features of the ith channel.

For some function abbreviations, TE denotes the transformer encoder layer, $\mathrm{RCons}_{k\times k}$ denotes the reconstruction to $k \times k$, $Up_{H\times W}$ means the upsampling to $H \times W$ by a bilinear interpolation, $\mathrm{Conv}_{1\times 1}$ denotes a learnable 1×1 convolutional layer, and σ is the sigmoid function. There is no doubt that the MHA in the transformer brings more computational complexity. To improve the computational efficiency, we abandon the original 8 or 16 heads in favor of fewer heads and less depth. Through experiments, we find that using a single head and one depth for MHA works best and greatly reduces the computational complexity. The details are shown in the ablation experiments.

The proposed TAM effectively enhances the representation of dense objects on the feature map by learning the interdependencies among channels through the original feature map, and then reconstructing the dense attention on spatial locations through the weights learned by the transformer.

D. Feature Refinement Module

The traditional FPN is used to process multiscale features, which can tackle the problem of scale variation in the objects. However, for remote sensing images, the scale variations in the

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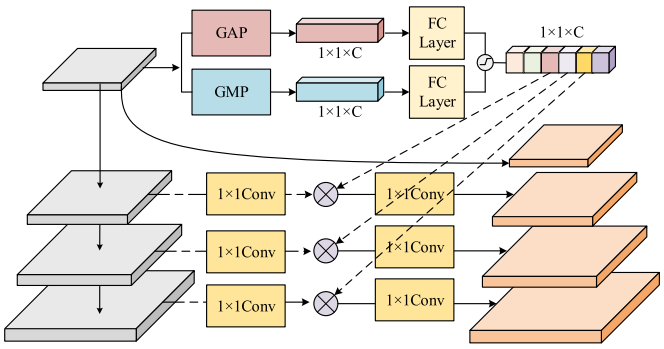


Fig. 5. Illustration of our proposed FRM.

objects are more drastic than those in natural scenes due to the inconsistent height of the bird's-eye view. In addition, the traditional pyramid's top-down propagation path of semantic information through the top level will involve a considerable information loss, and a large semantic gap will be resulted among the feature pyramid levels in remote sensing images. We introduce a new module in the feature pyramid, named FRM, to reduce the information loss by refining the top-level features to address the semantic gaps caused by scale variations.

As shown in Fig. 5, the semantic information of the top-level features of the FPN is refined by a dual-branch network module used for global information processing. On the top branch, we use GAP to extract location-insensitive semantic information. Different semantic information is assigned to different channels, and average pooling reduces the feature dimensions and captures consistent cues in the location-insensitive context. Therefore, consistent semantic information is lightly extracted among different channels by GAP on top-level features. Then an FC layer learns consistency weights to obtain the global semantics on the feature pyramid. For the down branch, we substitute global max pooling (GMP) for GAP. The GMP operation suffers from information loss compared with average pooling, but it can introduce the most salient information of the feature channel into the feature learning process. We apply GMP on the top-level features of the pyramid to learn the object saliency semantics with the aim of finding the maximum activation value within the receptive fields. This information helps alleviate the noise and semantic gaps in FRM. To eliminate the GMP loss, we fuse the weights learned by GMP and GAP and regularize them with a sigmoid function. The weights are calculated as follows:

$$\mathrm{FRM}(x) = \sigma\left(\mathrm{fc}(\mathrm{GAP}(x)) + \mathrm{fc}(\mathrm{GMP}(x))\right) \tag{9}$$

where σ represents the sigmoid function. To propagate the information learned by FRM to each level, we map the features of each level by 1×1 convolution, then multiply the weights of FRM to each level, and finally output the integrated feature pyramid by one another 1×1 convolution. The final output pyramid $[P_2, P_3, P_4, P_5]$ is calculated as follows:

$$P_i = \begin{cases} \mathrm{Conv}_{1\times1}(\mathrm{Conv}_{1\times1}(P_i) \otimes \mathrm{FRM}(P_5)) & \text{if} \quad i \neq 5 \\ P_i & \text{else} \end{cases} \tag{10}$$

where $\mathrm{Conv}_{1\times1}$ means the 1×1 convolutional layer, i represents the ith-level of pyramid, and $\otimes$ represents the elementwise multiplication. The proposed FRM effectively addresses object scale variation in remote sensing images by semantic guidance, alleviates the semantic gap caused by the scale at different levels, and it is a plug-and-play model.

E. Correlation Transformer Module

Inspired by image captioning, the input of this module is detected features of RoIs, which are then encoded by serialization, and the image caption is outputted by the decoder. The process from image to caption is similar to the process we use for object detection classification: encoding information from multiple object features and decoding semantics to identify their categories. The transformer is proposed to improve the performance of the image captioning model. Therefore, we use the thought of captioning and introduce the transformer to capture the dependencies among multiple objects. The CTM is proposed for better object classification in the classification branch of the head network.

For massive proposals generated by the RPN network, feature extraction is performed on these candidate regions using RoIAlign to obtain a large number of RoI features with the same scale. As shown in Fig. 6, to construct the set of RoI features to complete the training of the second stage network, k RoI features $[X_1, X_2, X_3, \ldots, X_k] \in \mathbb{R}^{C\times H\times W}$ generated by the RPN network are concatenated to be $T \in \mathbb{R}^{k\times(C\times H\times W)}$ feeding into the two branches of the head network. In the regression branch, similar to other methods, k is considered as the batch size, and the location offsets of the dense object regions are output through two FC layers separately as follows:

$$T = \mathrm{Concat}([X_1, X_2, X_3, \ldots, X_k]) \tag{11}$$

$$\mathrm{reg}(T) = \mathrm{fc}_2(\mathrm{fc}_1(T)) \tag{12}$$

where X_i represents the ith RoI, fc_1 and fc_2 represent the first FC layer and the second FC layer of the regression branch, respectively, and Concat represents the concatenation operation. In the classification branch, feature extraction is performed through one layer of FC layers to unify the feature space of dense regions. The proposed method fuses the position information using an object position encoder, and then inputs the fused features of k regions into the transformer encoder to learn the feature correlation among dense objects with the intention of extracting the association among different object categories. The features input to the classification branch are still the concatenated features T. Because the transformer inputs are used in a concatenated manner to construct sequences containing k embeddings and the computation is simplified by matrix multiplication during transformer computation, which treats different RoIs separately. The transformer outputs a feature $F \in \mathbb{R}^{k\times(C\times H\times W)}$, and this feature is fused with the features output by the first FC layer and features extracted by the regression branch. The coordinated feature information is finally transmitted through the last FC layer to output the classification confidence. The CTM details are calculated as follows:

$$F(x) = \mathrm{TE}(\mathrm{fusion}(\mathrm{fc}_3(x), \mathrm{ope}(x))) \tag{13}$$

$$\mathrm{CTM}(T) = \mathrm{fc}_4(\mathrm{fc}_3(T) + F(T) + \mathrm{fc}_1(T)) \tag{14}$$

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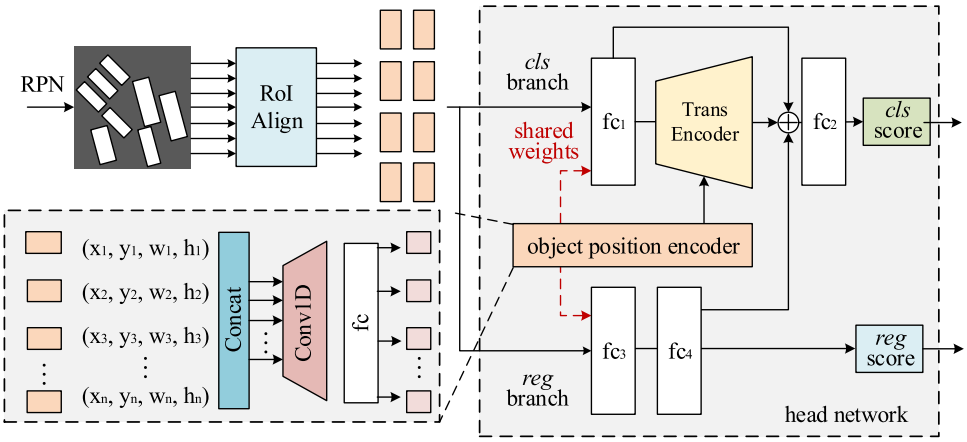


Fig. 6. Illustration of our proposed CTM with the object position encoder.

where TE represents the transformer encoder layer, fusion means the fusion function, fc_3 and fc_4 represent the first FC layer and the second FC layer of classification layer, respectively, and ope means the object position encoder. In particular, as mentioned in [61], there exists relevance between the classification and regression tasks, hence setting the first FC layer of both the branches, fc_1 and fc_3, to share parameters and increase the interactivity between classification and localization in CTM. Naturally, we also simplify the computation of MHA of the transformer in CTM. To reduce the computational complexity, we reduce the dimensionality of Q, K, and V vectors in the MHA and the dimensionality of the linear layers. The details are shown in the ablation experiments.

Compared with some methods in image captioning that discard the encoding of relative positions between object features, the proposed CTM considers that the position relationship of the objects is also useful to alleviate the classification difficulties of dense objects. As shown in Fig. 6, we propose a new position encoder for multiobject regions in CTM, which encodes the position information of each object by its position coordinates (x_i, y_i) as well as its width w and height h. For multiple objects $[o_1, o_2, o_3, \ldots, o_k]$ in dense scenes, RoI features can only represent their salient semantic information, and to introduce additional position information P_i, we represent their position vectors by a 1-D vector (x_i, y_i, w_i, h_i). In our proposed object position encoder, these vectors are fed through a layer of 1-D convolution (Conv1-D) to obtain the local correlation of the dense objects, and then through a layer of FC layers to obtain global correlation. The output of the final position encoding can be calculated as follows:

$$P_i = \mathrm{fc}(\mathrm{Conv1D}([x_i, y_i, w_i, h_i])) \tag{15}$$

where Conv1D represents the Conv1-D operation. To represent the object o_i, the location information P_i and the RoI feature X_i are fused as follows:

$$\mathrm{Fusion}(P_i, X_i) = \frac{P_i + X_i}{2}. \tag{16}$$

After CTM, the fused representation can effectively use the spatial and class relationships among the dense objects to assist the model to determine the class of independent objects better. In the regression branch, we adopt the Gliding Vertex [7] to represent the OBBs better. The loss function of our algorithm remains consistent with Gliding Vertex and is not redundantly described here. CTM is well-designed for head network, and its effective utility is also verified through subsequent experiments.

IV. Experiments

In this section, the experiments are conducted on two available OBB-based remote sensing image datasets to validate the effectiveness of the proposed CLT-Det and further demonstrate our detection performance. First, we elaborate on the information of datasets, evaluation metrics, and implementation details. Then, ablation experiments are performed to verify the effectiveness of individual components of CLT-Det, including TAM, FRM, and CTM. In addition, the effect of some hyperparameters on the model performance is investigated in the ablation experiments. Ultimately, we compare several recent OBB-based object detection methods on two datasets to demonstrate the advantage of our method. All the experiments are executed on a computer with Intel Xeon Gold 5120 CPU at 2.20 GHz and 4 NVIDIA GeForce RTX2080Ti GPU (11-GB memory) for acceleration.

A. Datasets

1) DOTA: DOTA is a benchmark dataset for remote sensing image object detection, which contains 2806 remote sensing images from different sensors. The images were acquired from Google Earth and the China Center for Resources Satellite Data and Application. The imagery size varies from 800 × 800 pixels to 4000 × 4000 pixels, which contains 188 282 instances with different scales and angular orientations. These instances are labeled into 15 categories: 1) plane (PL); 2) baseball diamond (BD); 3) bridge (BR); 4) ground track field (GTF); 5) small vehicle (SV); 6) large vehicle (LV); 7) ship (SH); 8) tennis court (TC); 9) basketball court (BC); 10) storage tank (ST); 11) soccer-ball field (SBF); 12) roundabout (RA); 13) HA; 14) swimming pool (SP); and 15) helicopter (HC).

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TABLE I
ABLATION STUDY OF CLT-DET ON THE DOTA DATASET (%)

	submodules			PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
	+ TAM	+ FRM	+ CTM																
baseline	-	-	-	**89.60**	84.67	49.54	74.14	71.64	71.29	86.66	90.80	80.18	86.16	58.26	67.58	66.65	69.86	59.97	73.80
	✓	-	-	89.44	84.58	51.44	**78.01**	72.17	72.80	87.36	90.83	80.05	86.70	59.02	68.72	72.30	71.34	57.52	74.82
	-	✓	-	89.56	85.00	52.04	73.75	73.47	73.64	87.00	90.83	80.37	86.75	58.50	69.29	74.96	71.88	59.42	75.10
	-	-	✓	89.33	**86.21**	52.79	73.15	71.69	72.82	87.37	90.84	79.56	**87.10**	58.82	68.16	74.25	71.80	56.54	74.70
CLT-Det	✓	✓	-	89.48	83.92	**54.35**	72.93	74.77	75.15	87.11	90.86	80.25	86.55	58.37	**71.33**	74.88	**72.40**	60.32	75.51
	✓	-	✓	89.53	85.15	52.21	75.59	77.41	77.34	88.27	90.88	**86.42**	86.85	62.35	69.99	**75.38**	70.12	65.55	76.87
	-	✓	✓	88.88	85.38	53.47	78.65	74.54	73.74	87.17	90.80	84.86	86.30	58.20	68.71	74.69	70.81	66.27	76.16
	✓	✓	✓	89.31	85.69	53.97	77.11	**79.66**	**79.01**	**88.55**	**90.89**	85.36	86.56	**63.92**	68.47	75.65	70.65	**66.91**	**77.45**

2) HRSC2016: The HRSC2016 is an SH detection remote sensing dataset with OBB-based annotations. It contains 1061 remote sensing images acquired from Google Earth. The image sizes range from 300 × 300 to 1599 × 900 and contain 2976 SH instances. In HRSC2016, 436 images are used as the training set, 181 as the validation set, and 444 as the test set.

B. Evaluation Metrics

To evaluate the detection accuracy of the proposed model, we adopt the mean average precision (mAP) in object detection as the evaluation metric. mAP is calculated by averaging the average precision (AP) values of each object class. For object detection, the larger the AP and mAP, the better the accuracy of the detection model. The formulation of mAP is calculated as follows:

$$\text{mAP} = \frac{1}{n}\sum_{i}^{n} \text{AP}_i \tag{17}$$

where i denotes the ith category, and n denotes the total number of categories. AP_i is the AP of the ith category. The value of AP is the area under the precision–recall ($P - R$) curve for each category. Precision expresses the proportion of detected objects that are true positives, while recall expresses the proportion of correctly detected positives. P and R are calculated as follows:

$$P = \frac{\text{TP}}{\text{FP} + \text{TP}} \tag{18}$$

$$R = \frac{\text{TP}}{\text{FP} + \text{FN}} \tag{19}$$

where TP, FP, and FN denote the numbers of true positives, false positives, and false negatives, accordingly.

C. Implementation Details

All the models in the experiments were implemented based on PyTorch. The backbone of CLT-Det is ResNet-101 pretrained on ImageNet and then fine-tuned on the remote sensing dataset. We construct our baseline model based on the Faster R-CNN with FPN, using Gliding Vertex to regress OBBs. We choose features from the last four stages of ResNet to construct the pyramid. The number of FPN channels is set to 256. During the training process, direct detection degrades the model performance due to the large resolution and varying sizes of the images in DOTA. Therefore, we split the images into 1024 × 1024 pixels subimages with 200 pixels overlap. In addition, we perform data enhancement on the images by rotating 90°, 180°, and 270°. Consistent with other state-of-the-art methods, we use multiscale training and testing with scales of [0.5, 1.0, 1.5]. For the training of HRSC2016, we do not crop the images and do not apply additional data augmentation methods, but only resize the images to 512 × 800.

For a fair comparison with other state-of-the-art methods, some hyperparameters need to be consistent in the experiments. In the experiments in DOTA, we set the initial learning rate to 7.5×10^{-3} and the final learning rate to 7.5×10^{-5}. The total number of training iterations is 70 000, and the batch size of each iteration is four. In the experiments in HRSC2016, the initial learning rate is 1×10^{-2}, and the final learning rate is 1×10^{-4}. The total iterations of HRSC2016 are 2000 with a batch size of eight. The weight decay is set to 1×10^{-4}, and the momentum is 9×10^{-1} for two datasets. We set the anchor boxes sizes of each layer as $[32^2, 64^2, 128^2, 256^2, 512^2]$ and the anchor box ratios to [0.5, 1.0, 2.0]. The NMS threshold is set to 0.5.

D. Ablation Study

In this section, ablation experiments are conducted to verify the effectiveness of each module in CLT-Det and the effect of some hyperparameters in each module on detection performance. The ablation experiments are performed on the DOTA dataset and the HRSC2016 dataset, respectively. The baseline model used in the experiments is Faster R-CNN with FPN, and the feature extraction network is ResNet-101. The OBB representation of Gliding Vertex is leveraged to achieve higher detection performance.

1) Effectiveness of TAM: The objects of remote sensing images suffer from the problem of feature extraction due to the limitation of view. The traditional backbone networks may affect the detection performance. Therefore, in CLT-Det, TAM is proposed to improve the saliency representation of objects.

To verify the effectiveness of TAM, we designed ablation experiments on two datasets, DOTA and HRSC2016, with ResNet-101 as the feature extraction network, while TAM is introduced in ResNet for comparison. Both the AP of a single class and mAP are used as metrics to evaluate TAM, and the experimental results are shown in Tables I and II.

Our ablation experiments of the DOTA dataset are shown in Table I, where line 2 shows the effect of the performance when

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TABLE II
ABLATION STUDY OF CLT-DET ON THE HRSC2016 DATASET (%)

	submodules			mAP
	+ TAM	+ FRM	+ CTM	
baseline	-	-	-	88.59
CLT-Det	✓	-	-	89.18
	-	✓	-	89.00
	-	-	✓	89.31
	✓	✓	-	89.01
	✓	-	✓	88.82
	-	✓	✓	89.57
	✓	✓	✓	**89.78**

TAM is introduced in the baseline. After introducing TAM on ResNet, the model shows significant improvement in detection performance for some categories of objects. The most obvious of them are as follows, the model improves 5.65% of AP on HA, 3.87% on GTF, and 1.90% on BR. Since HA and BR are too narrow in remote sensing images and occupy fewer pixels, their features are blurred. GTF often appears with complex scenes and is easily obscured by other objects so that their features are more difficult to extract. The experimental results demonstrate that the detection performance of these categories can be effectively improved through enhancement of representation by TAM. In addition, some objects with smaller sizes, such as vehicles and ships, have a small improvement (about 0.5% AP). Eventually, introducing TAM in ResNet effectively boosts the model mAP of 1.05% on the DOTA dataset.

The ablation experiment to validate the effectiveness of TAM on the HRSC2016 dataset is presented in line 2 of Table II. For the detection of ships with relatively large aspect and small size, TAM is able to effectively improve their feature representation. The experimental results demonstrate that the introduction of TAM improves the detection mAP by 0.59%.

2) Effectiveness of FRM: Due to more drastic variations in object scales in remote sensing images, the traditional multiscale detection approach generates a large semantic gap. To solve this issue, FRM introduced in CLT-Det is aimed to guide the propagation of top-level semantic information downward. We devise ablation experiments on DOTA and HRSC2016 to verify the effectiveness of FRM for scale variation issues. In the experiments, we improve the traditional four-layer FPN. The results of the improved FPN are shown in Tables I and II.

Line 3 of Table I demonstrates the ablation experiments of validating the FRM on the DOTA dataset. With the introduction of FRM on the four-layer feature pyramid, the detection performance is effectively improved on several categories with a large range of scale variations. The most obvious improvement remains in the HA category. We consider that in the DOTA dataset, the HA category has a large-scale variation and is difficult to distinguish due to its proximity to the coast, which can easily lead to ambiguity, so FRM can effectively solve the semantic information gap. In addition, FRM has promising performance in categories such as vehicles (SV, LV) and BDs, which can indicate the excellent performance of FRM on small and multiscale objects. Ultimately, compared with the baseline model, FRM effectively improves the mAP by 1.3% on the DOTA dataset.

For the HRSC2016 dataset, line 3 of Table II demonstrates the effectiveness of FRM in SH detection. The HRSC2016 dataset contains more SH objects of different scales, so it is verified that FRM can effectively improve the mAP by 0.41%.

3) Effectiveness of CTM: For remote sensing image object detection, the accurate recognition of dense objects is one of the major challenges. To better identify objects in dense scenes, correlation learning among the objects is essential. The CTM proposed in this article uses transformer to model the correlation among objects in the image to assist the classification of dense objects. To verify the effectiveness of CTM, we elaborately design ablation experiments to introduce the CTM in the head network and compare the changes in performance before and after. The results of the ablation experiments for DOTA and HRSC2016 are shown in Tables I and II.

As shown in line 3 of Table I, the total detection performance after the introduction of CTM increased by 0.9% from the baseline model. In particular, CTM has more outstanding performance in the categories of BD, ST, and BR, with AP values of 86.21%, 87.10%, and 52.79%, respectively. Undoubtedly, these objects tend to appear in more complex scenes, for example, BRs appearing simultaneously with ships and HA, and therefore, when CTM is introduced, the inspection accuracy of BRs, ships, and ports is improved accordingly. CTM learns the correlation among these objects to improve the detection performance.

As shown in the third row of Table II, for the SH detection task in HRSC2016, CTM can significantly improve the mAP by 0.72%. For single-class detection, there are also learnable correlations among different ships, for example, when there is one SH infested, there are often multiple ships infested nearby as well.

Therefore, capturing the correlation among multiple instances of single class using CTM can also improve the performance of single-class detection. As shown in Fig. 7, with the introduction of CTM, the correlation information can be used to reinforce the features of those objects that are difficult to detect.

In addition, to explore the trade-off between detection performance and efficiency, we ablate the linear layer of four involved in the CTM to obtain a balance between performance and efficiency. The experimental results are shown in Fig. 8.

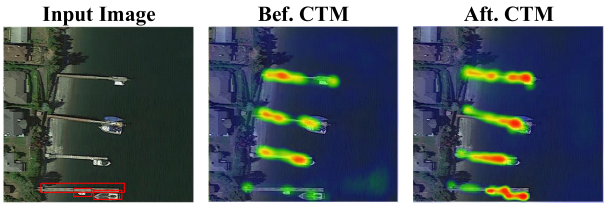


Fig. 7. Visualization of feature maps produced by models with CTM. Both the feature maps before and after CTM are visualized. Red solid boxes indicate objects that are difficult to detect due to their dense distribution, small size, or at the edges of the image.

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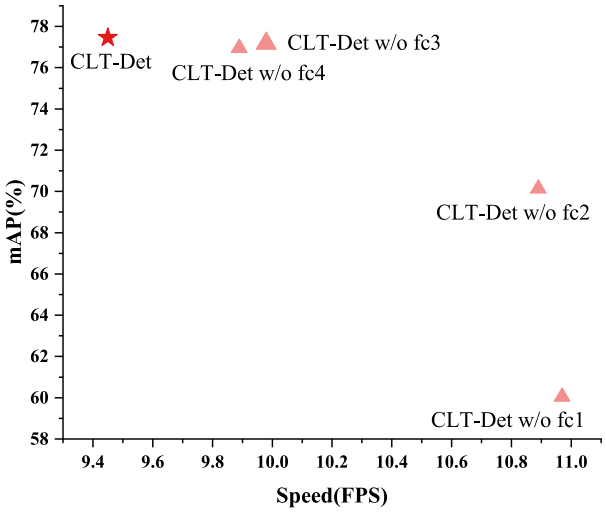


Fig. 8. Speed and accuracy comparison with ablation of models with different linear layers.

When all the linear layers are used, the mAP value is the highest, but the detection efficiency is the worst. Then we remove the linear layers fc1, fc2, fc3, and fc4, respectively. When fc1 and fc2 are removed, the detection performance decreases dramatically, while when fc3 and fc4 are removed, the detection performance decreases within an acceptable range and the detection efficiency is improved to a certain extent.

4) Combined Variants: In the other lines of Tables I and II, the effectiveness of the variants obtained by different combinations of TAM, FRM, and CTM are shown. In Table I, we can find that when TAM and FRN are combined, the mAP reaches 75.51 and does not change considerably. The possible reason is that TAM disturbs the information of the top-level features, causing the semantic information learned by FRM to be inaccurate. In contrast, the TAM and CTM combination and the combination of FRM and CTM improve the mAP by 3.07% and 2.36% on the baseline model, respectively. Finally, fusing the three modules to obtain CLT-Det effectively boosted the mAP by 3.65% on the DOTA dataset.

The results on the HRSC2016 data are similar to DOTA in Table II, where the direct combination of TAM and FRM does not bring the desired performance. In addition, the combination of TAM and CTM differently brings some performance degradation. The possible reason is that the HRSC2016 dataset is only a single category, and it is difficult for TAM and CTM to capture the global information effectively; instead, some interference factors are introduced unnecessarily to affect the detection performance. Undoubtedly, when combining TAM, FRM, and CTM to obtain CLT-Det, we improved the mAP on HRSC2016 by 1.19% and obtained the best detection performance of 89.78%.

5) Impact of Hyperparameters: Here, we devise several sets of ablation experiments to validate the optimal hyperparameter settings to obtain better detection performance of the model. We find that TAM and CTM are more sensitive to hyperparameters due to the transformer. For the transformer in TAM,

TABLE III
INFLUENCE (%) OF THE NUMBER OF HEADS AND DEPTH ON TAM

Model		Head	Depth	mAP
Baseline	ResNet	0	0	73.80
ResNet with TAM		1	1	**74.82**
		2	1	73.54
		1	2	74.03

the module is sensitive to the number of heads and MHA in the transformer encoder. In addition, the performance of TAM is also related to the feature scale obtained during GAP. While for CTM, its performance is more impacted by the dimensions of the embedding and linear layers in the encoder. Moreover, CTM is more sensitive to the branch that is introduced in the head network. The learning method in the object position encoder proposed in CTM also determines the stability of its performance. The baseline model used in the hyperparameter experiments of TAM is Faster R-CNN with FPN, while the baseline model used in the ablation experiments of CTM is the model proposed in this article after the introduction of TAM. All the ablation experiments are performed on the DOTA dataset.

Effect of the number of heads and depth of TAM. The transformer uses an MHA mechanism to learn semantic information in different feature spaces. We try different numbers of heads and depths to explore the optimal settings. As shown in Table III, when we use one head and one depth of the transformer, we obtained an mAP of 74.82%, which is an increase of 1.08% compared with the baseline. Then we keep the number of layers constant and increase the number of heads to two, and the model performance drops to 73.54%. When we keep the number of layers at one and increase the number of heads to two, the model performance decreases to 74.03%. Thus, the model achieves optimal performance when the number of heads is one and the depth is one.

Effect of the pooling size of TAM. In TAM, the purpose of GAP is to obtain global information and reduce the dimensions of the transformer input. Intuitively, if the pooling size is too small, it may lead to a substantial loss of detailed features, while a large pooling size may cause difficulties in the computation of the transformer. In the ablation experiments, we set the pooling sizes to (4, 4), (8, 8), (16, 16), and (32, 32) to explore the optimal parameters. As shown in Fig. 9, different pooling sizes have different advantages and disadvantages on categories. For (4, 4), the detection is more accurate at larger scale objects such as RA and BRs. For (8, 8), the detection is better for more dense and variable scale objects such as SVs and boats (SH). For (16, 16) and (32, 32), the detection effect is more different on categories, concentrating on the detection of objects with larger scales. By comparing mAP, we choose (8, 8) as the pooling size of GAP to obtain better detection performance.

Effect of the dimensions of embedding and linear layers of CTM. In CTM, we use the transformer settings to maintain the same pattern as in image captioning. During the experiments, we find that different dimensions of embedding and dimen-

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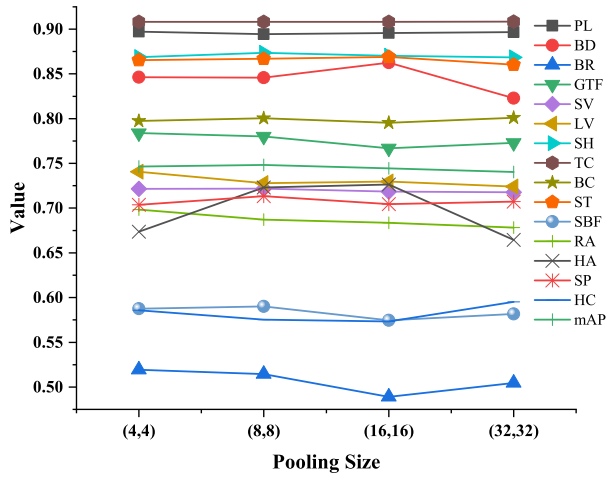


Fig. 9. Illustration of the pooling size for validation of GAP in TAM on the DOTA dataset. Comparison with (4, 4), (8, 8), (16, 16), and (32, 32) AP values in 15 categories and validation of the effect of pooling size on model performance by mAP values.

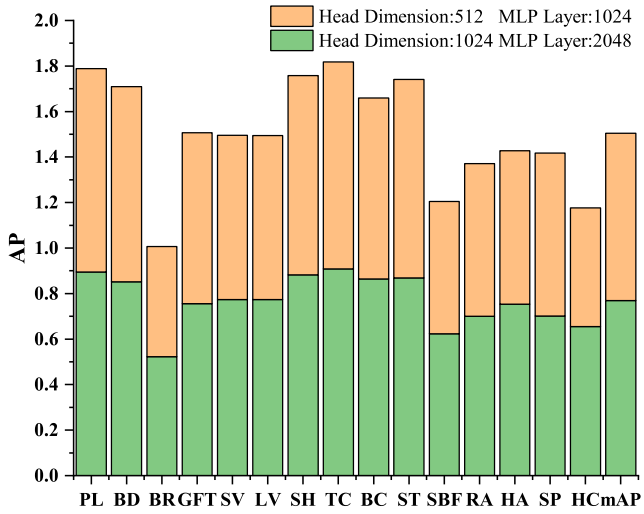


Fig. 10. Illustration of the effect on CTM for different embedding dimensions and FC layer dimensions. Comparison with mAP for embedding dimensions and FC layer dimensions of 512, 1024 and 1024, 2048, respectively.

sions of MLP affect the detection performance. We explore the performance comparison of two settings commonly used in image captioning. As shown in Fig. 10, setting the embedding dimension to 512 and the FC layer dimension to 1024 performs better on multiple categories, and mAP works better than setting the dimension to 1024, 2048.

Effect of the learning ways of the object position encoder. Our proposed object position encoder in CTM can work with an FC layer or a Conv1-D for learning the relative position relationships among multiple objects. Intuitively FC focuses on learning global dependencies, while Conv1D is adept at mining the correlations among local objects. As shown in Table IV, CLT-Det achieves an mAP value of 75.14% when using the FC layer and 76.87% when using Conv1-D. Therefore, correlation among local objects in remote sensing images is more critical for classification.

TABLE IV
INFLUENCE (%) OF THE LEARNING WAYS OF OBJECT POSITION ENCODER ON CTM

Model		Position information	mAP
Baseline	CLT-Det without CTM	-	74.82
CLT-Det with CTM		-	74.80
		FC	75.14
		Conv1D	**76.87**

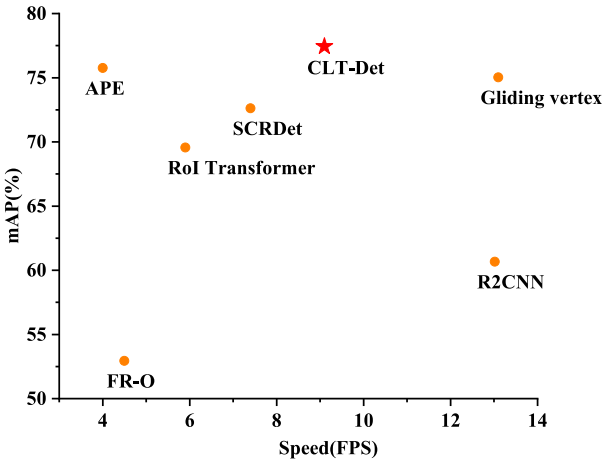


Fig. 11. Speed and accuracy comparison with several state-of-the-art methods.

TABLE V
INFLUENCE (%) OF BRANCH IN HEAD NETWORK

Model		*reg* Branch	*cls* Branch	mAP
Baseline	CLT-Det without CTM	-	-	74.82
CLT-Det with CTM		✓	-	73.75
		-	✓	**76.87**
		✓	✓	73.70

TABLE VI
INFLUENCE (%) OF LAYER OF FEATURES ON FRM

Model	P_2	P_3	P_4	P_5	mAP
FRM	-	-	-	-	73.80
	✓	-	-	-	71.14
	-	✓	-	-	72.27
	-	-	✓	-	73.72
	-	-	-	✓	**75.10**

Effect of the branch in the head network. The head network is divided into regression branches and classification branches. The regression branch is responsible for predicting the position coordinates of the OBBs, while the classification branch predicts the category of the objects in the OBBs. We explore the impact of the proposed CTM on the classification task and the regression task. As shown in Table V, the model has a large performance degradation when we introduce the object correlation learned by the CTM into the regression branch. The model performance is also weak when the correlations are

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TABLE VII
ACCURACY (%) COMPARISON WITH THE STATE-OF-THE-ART METHODS ON THE DOTA DATASET

Method	Backbone	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
FR-O [45]	ResNet-50	79.42	77.13	17.70	64.05	35.30	38.021	37.16	89.41	69.64	59.28	50.30	52.91	47.89	47.40	46.30	54.13
PIoU Loss [62]	DLA-34	80.90	69.70	24.10	60.20	38.30	64.40	64.80	90.90	77.20	70.40	46.50	37.10	57.10	61.90	64.00	60.50
RRPN [54]	ResNet-101	88.52	71.20	31.66	59.30	51.85	56.19	57.25	90.81	72.84	67.38	56.69	52.84	53.08	51.94	53.58	61.01
RoI Transformer [6]	ResNet-101	88.64	78.52	43.44	75.92	68.81	73.68	83.59	90.74	77.27	81.46	58.39	53.54	62.83	58.93	47.67	69.56
CAD-Net [55]	ResNet-101	87.80	82.40	49.40	73.50	71.10	63.50	76.70	90.90	79.20	73.30	48.40	60.90	62.00	67.00	62.20	69.90
DRN [63]	Hourglass-104	89.71	82.34	47.22	64.10	76.22	74.43	85.84	90.57	86.18	84.89	57.65	61.93	69.30	69.63	58.48	73.23
R^3Det [17]	ResNet-101	88.76	83.09	50.91	67.27	76.23	80.39	86.72	90.78	84.68	83.24	61.98	61.35	66.91	70.63	53.94	73.79
RSDet [14]	ResNet-152	**90.10**	82.00	53.80	68.50	70.20	78.70	73.60	**91.20**	87.10	84.70	64.30	68.20	66.10	69.30	63.70	74.10
Gliding Vertex [7]	ResNet-101	89.64	85.00	52.26	77.34	73.01	73.14	86.82	90.74	79.02	86.81	59.55	**70.91**	72.94	70.86	57.32	75.02
Zand at.el. [8]	DarkNet-53	89.90	80.40	50.50	76.00	78.20	77.80	87.40	90.00	83.20	86.93	63.30	68.70	67.60	70.80	63.80	75.50
CBDA-Net [9]	DLA-34	89.17	85.92	50.28	65.02	77.72	82.32	87.89	90.48	86.47	85.90	66.85	66.48	67.41	71.33	62.89	75.74
APE [57]	ResNeXt-101	89.96	83.62	53.42	76.03	74.01	77.16	79.45	90.83	87.15	84.51	67.72	60.33	74.61	71.84	65.55	75.75
CenterMap OBB [10]	ResNet-101	89.83	84.41	54.60	70.25	77.66	78.32	87.19	90.66	84.89	85.27	56.46	69.23	74.13	71.56	66.06	76.03
MRDet [64]	ResNet-101	89.49	84.29	55.40	66.68	76.27	82.13	87.86	90.81	86.92	85.00	52.34	65.98	76.22	76.78	67.49	76.24
OPLD [15]	ResNet-101	89.37	85.82	54.10	**79.58**	75.00	75.13	86.92	90.88	86.42	86.62	62.46	68.41	73.98	68.11	63.69	76.43
Polar Ray [65]	ResNet-101	88.78	85.51	51.89	73.42	74.36	77.98	87.78	90.86	86.50	86.84	60.69	69.84	75.29	72.22	67.22	76.50
Constrained-SIoU [66]	ResNet-50	89.39	84.00	54.76	73.71	76.22	78.31	83.73	90.87	84.96	87.37	62.38	67.15	73.86	69.44	**71.98**	76.54
PolarDet [13]	ResNet-101	89.65	**87.07**	48.14	70.97	78.53	80.34	87.45	90.76	85.63	86.87	61.64	70.32	71.92	73.09	67.15	76.64
CFA [67]	ResNet-152	89.08	83.20	54.37	66.87	**81.23**	80.96	87.17	90.21	84.32	86.09	52.34	69.94	75.52	**80.76**	67.96	76.67
SCRDet++ [68]	ResNet-101	90.05	84.39	**55.44**	73.99	77.54	71.11	86.05	90.67	87.32	87.08	**69.62**	68.90	73.74	71.29	65.08	76.81
DAL [69]	ResNet-50	89.69	83.11	55.03	71.00	78.30	81.90	88.46	90.89	84.97	**87.46**	64.41	65.65	**76.86**	72.09	64.35	76.95
GGHL [70]	DarkNet-53	89.74	85.63	44.50	77.48	76.72	80.45	86.16	90.83	**88.18**	86.26	67.07	69.40	73.38	68.45	70.14	76.95
DCL [58]	ResNet-152	89.26	83.60	53.54	72.76	79.04	**82.56**	87.31	90.67	86.59	86.98	67.49	66.88	73.29	70.56	69.99	77.37
CLT-Det(ours)	**ResNet-101**	89.31	85.69	53.97	77.11	79.66	79.01	**88.55**	90.89	85.36	86.56	63.92	68.47	75.65	70.65	66.91	**77.45**

TABLE VIII
ACCURACY (%) COMPARISON WITH THE STATE-OF-THE-ART METHODS ON THE HRSC2016 DATASET

Method	Backbone	Input size	mAP
R^2CNN [51]	ResNet-101	800×800	79.73
RRPN [54]	ResNet-101	800×800	79.08
TOSO [71]	ResNet-101	800×800	79.29
RoI Transformer [6]	ResNet-101	512×800	86.20
RSDet [14]	ResNet-50	800×800	86.50
Gliding Vertex [7]	ResNet-101	512×800	88.20
DAL [69]	ResNet-101	416×416	88.95
R^3Det [17]	ResNet-101	800×800	89.26
DCL [58]	ResNet-101	800×800	89.46
CSL [16]	ResNet-50	800×800	89.62
CLT-Det(ours)	**ResNet-101**	**512×800**	**89.72**

introduced into both the classification and regression branches. Only when CTM is introduced into the classification branch, the model has an expected performance improvement with an mAP of 76.87%. Therefore, it can be concluded that the learned correlations from CTM help in predicting the object categories.

Effect of the layer of features on FRM. The size of features varies in different layers of feature extraction stage, and the richness of semantic information contained in the features varies. We explore which layer is used for semantic guidance in FRM. As shown in Table VI, semantic guidance is least effective when using the P_2 layer, which is due to the fact that although the P_2 layer contains most of the original features of the objects, there is too little semantic information to guide the learning of rich semantics at a higher level. The best results are obtained when P_5 is used, because P_5 contains rich semantic information despite its small size and can guide the learning of richer target representations at higher levels.

E. Comparisons With State-of-the-Art

1) Results on DOTA: The DOTA dataset contains 15 categories of remote sensing image ground objects. As shown in Table VII, we compare our method with some state-of-the-art detectors. The baseline model we use is a combination of oriented faster region-CNN (FR-O) [45] and Gliding Vertex. After improvement, CLT-Det achieves 77.45% on the DOTA dataset, surpassing many current advanced methods such as SCRDet++ [68], DCL [58], gaussian heatmap label assignment (GGHL) [70], and dynamic anchor learning (DAL) [69]. For a fair comparison, our method and CenterMap OBB [10] and APE [57] both use the two-stage object detector for improvement, and our method surpasses them with the simultaneous use of ResNet-101 0.78% and 1.03% of mAP. In addition, we compare the speed advantages and disadvantages of CLT-Det with other methods. As shown in Fig. 11, compared with Gliding Vertex, R^2CNN [51], CLT-Det is slower but has greater accuracy improvement. Therefore, in this article, we propose CLT-Det to exchange a certain speed cost for the superiority in accuracy. The results of visualization on the DOTA dataset are shown in Fig. 12.

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Fig. 12. Visualization of detection results on the DOTA dataset.

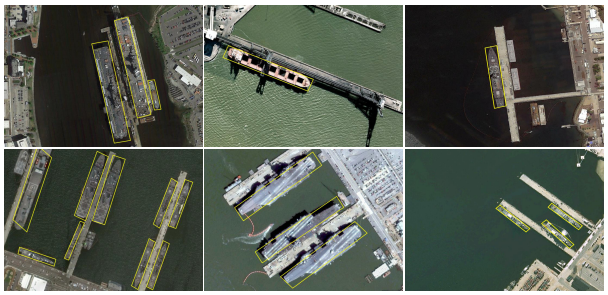


Fig. 13. Visualization of detection results on the HRSC2016 dataset.

2) Results on HRSC2016: The HRSC2016 dataset contains a large number of oriented ships. As shown in Table VIII, many methods attain good performance in this dataset, such as DAL [69], R^3Det [17], DCL [58], and CSL [16]. CLT-Det proposed in this article uses ResNet-101 as the feature extraction network with an input image size of 512×800 and achieves an mAP value of 89.72% surpassing several of the superior methods mentioned above. In addition, we present some visualization results on the HRSC2016 test set in Fig. 13.

V. Conclusion

In this article, a two-stage object detection method, i.e., CLT-Det, is proposed for detecting dense objects in remote sensing images. In the feature extraction stage, the interdependency mining of dense object features effectively enhances the representation capability of the model for dense objects by adopting the TAM. In the localization and classification stage, the correlation among multiple objects is learned by the CTM to effectively enhance the model's classification for dense objects. Besides, the FRM is proposed in the FPN to deal with the issue of drastic variations in the scale of objects using the semantic information of the top-level features that guide the learning of the bottom-level features. The effectiveness of the proposed method is validated on two popular used datasets. In the future, we will devote ourselves to optimizing the representation of OBBs to enhance the representation capability of the model for rotation-invariant features.

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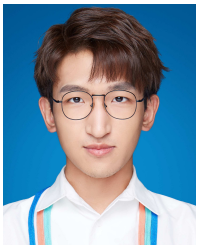
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